

Quiz 6 Answers

Phil 305

$\Box(A \vee B) \rightarrow (\Diamond A \vee \Box B)$ is a theorem of KT

1.	$\mathbb{F} 1, \Box(A \vee B) \rightarrow (\Diamond A \vee \Box B) \checkmark$	Assumption
2.	$\mathbb{T} 1, \Box(A \vee B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond A \vee \Box B \checkmark$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{F} 1, \Diamond A$	$\vee \mathbb{F} 3$
5.	$\mathbb{F} 1, \Box B \checkmark$	$\vee \mathbb{F} 1$
6.	$\mathbb{F} 1.1, B$	$\Box \mathbb{F} 5$
7.	$\mathbb{T} 1.1, A \vee B \checkmark$	$\Box \mathbb{T} 2$
└───┬───┘		
8.	$\mathbb{T} 1.1, A$	$\vee \mathbb{T} 7$
9.	$\mathbb{F} 1.1, A$	$\Diamond \mathbb{F} 4$
	$\otimes$	

$\Box(A \rightarrow B) \rightarrow (A \rightarrow \Box B)$ is not a theorem of KT

1.	$\mathbb{F} 1, \Box(A \rightarrow B) \rightarrow (A \rightarrow \Box B) \checkmark$	Assumption
2.	$\mathbb{T} 1, \Box(A \rightarrow B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, A \rightarrow \Box B \checkmark$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, A$	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Box B$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{T} 1, A \rightarrow B \checkmark$	$\mathbb{T} \Box 2$
└───┬───┘		
7.	$\mathbb{F} 1, A$	$\rightarrow \mathbb{T} 6$
8.	$\otimes$	$\Box \mathbb{F} 5$
	$\mathbb{T} 1, B$	
9.	$\mathbb{F} 1.1, B$	$\Box \mathbb{T} 2$
	$\mathbb{T} 1.1, A \rightarrow B \checkmark$	
└───┬───┘		
10.	$\mathbb{F} 1.1, A$	$\rightarrow \mathbb{T} 9$
	$\mathbb{T} 1.1, B$	
	$\otimes$	

The middle branch stays open, and the open-ended instruction at line 2 has been applied everywhere that's needed. The counterexample is where A and B are both true at w_1 and both false at $w_{1.1}$.

$\Box A \rightarrow (A \wedge \Box \Box A)$ is not a theorem of K4

1.	$\mathbb{F} 1, \Box A \rightarrow (A \wedge \Box \Box A) \checkmark$	Assumption
2.	$\mathbb{T} 1, \Box A$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, A \wedge \Box \Box A \checkmark$	$\rightarrow \mathbb{F} 1$
└───┬───┘		
4.	$\mathbb{F} 1, A$	$\wedge \mathbb{F} 3$
5.	$\mathbb{F} 1, \Box \Box A \checkmark$	$\Box \mathbb{F} 4$
6.	$\mathbb{F} 1.1, \Box A$	$4 \Box 2$
	$\mathbb{T} 1.1, \Box A$	
	$\otimes$	

The left hand branch stays open. The counterexample is the somewhat degenerate model where w_1 can't access anything, and has A false.

$\Diamond(A \rightarrow \Diamond B) \rightarrow (\Box A \rightarrow \Diamond(A \wedge B))$ is a theorem of K4

1.	$\mathbb{F} 1, \Diamond(A \rightarrow \Diamond B) \rightarrow (\Box A \rightarrow \Diamond(A \wedge B))$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond(A \rightarrow \Diamond B)$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box A \rightarrow \Diamond(A \wedge B)$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Box A$	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond(A \wedge B)$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{T} 1.1, A \rightarrow \Diamond B$ ✓	$\Diamond \mathbb{T} 2$
7.	$\mathbb{T} 1.1, A$	$\Box \mathbb{T} 4$
8.	$\mathbb{F} 1.1, A \wedge B$	$\Diamond \mathbb{F} 5$
└───┬───		
9.	$\mathbb{F} 1.1, A$ $\mathbb{T} 1.1, \Diamond B$ ✓	$\rightarrow \mathbb{T} 6$
10.	⊗ $\mathbb{T} 1.1, \Box A$	$4\Box 4$
11.	$\mathbb{F} 1.1, \Diamond(A \wedge B)$	$4\Diamond 5$
12.	$\mathbb{T} 1.1.1, B$	$\Diamond \mathbb{T} 9$
13.	$\mathbb{T} 1.1.1, A$	$\Box \mathbb{T} 10$
14.	$\mathbb{F} 1.1.1, A \wedge B$ ✓	$\Diamond \mathbb{F} 11$
└───┬───		
15.	$\mathbb{F} 1.1.1, A$ $\mathbb{F} 1.1.1, B$	$\wedge \mathbb{F} 14$
	⊗ ⊗	

$\Diamond(A \wedge B) \rightarrow \Box \Diamond(C \rightarrow B)$ is not a theorem of S4

1.	$\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow \Box \Diamond(C \rightarrow B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond(A \wedge B)$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box \Diamond(C \rightarrow B)$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1.1, A \wedge B$ ✓	$\Diamond \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\wedge \mathbb{T} 4$
6.	$\mathbb{T} 1.1, B$	$\wedge \mathbb{T} 4$
7.	$\mathbb{F} 1.2, \Diamond(C \rightarrow B)$	$\Box \mathbb{F} 3$
8.	$\mathbb{F} 1.2, C \rightarrow B$ ✓	$\mathbb{T} \Diamond 7$
9.	$\mathbb{T} 1.2, C$	$\rightarrow \mathbb{F} 8$
10.	$\mathbb{F} 1.2, B$	$\rightarrow \mathbb{F} 8$

The trunk stays open, and we've applied the open-ended rule everywhere it needs to be applied. The countermodel has w_1 , which can access $w_{1.1}$ and $w_{1.2}$. At $w_{1.1}$ both A and B are true. At $w_{1.2}$, C is true and B is false. Any way of filling in the other values will still create a counterexample.

$\Diamond\Diamond A \rightarrow (\Diamond A \wedge \Diamond\Diamond\Diamond A)$ is a theorem of S4

1.	$\mathbb{F} 1, \Diamond\Diamond A \rightarrow (\Diamond A \wedge \Diamond\Diamond\Diamond A)$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond\Diamond A$ ✓	$\rightarrow\mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond A \wedge \Diamond\Diamond\Diamond A$ ✓	$\rightarrow\mathbb{F} 1$
4.	$\mathbb{T} 1.1, \Diamond A$	$\Diamond\mathbb{T} 2$
└───┬───┘		
5.	$\mathbb{F} 1, \Diamond A$	$\wedge\mathbb{F} 3$
6.	$\mathbb{F} 1.1, \Diamond A$	$4\Diamond 5$
7.	$\mathbb{F} 1, \Diamond\Diamond\Diamond A$	$\mathbb{T}\Diamond 6$
	$\otimes$	$\otimes$

There are a lot of ways to close this one; this is the quickest I found but it's definitely not the only one.

$\Diamond\Box A \rightarrow \Diamond A$ is a theorem of KTB

1.	$\mathbb{F} 1, \Diamond\Box A \rightarrow \Diamond A$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond\Box A$ ✓	$\rightarrow\mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond A$	$\rightarrow\mathbb{F} 1$
4.	$\mathbb{T} 1.1, \Box A$	$\Diamond\mathbb{T} 2$
5.	$\mathbb{T} 1, A$	$\Box\mathbb{T} 4$
6.	$\mathbb{F} 1, A$	$\mathbb{T}\Diamond 3$
	$\otimes$	

$(\Box A \wedge \Box B) \rightarrow \Box\Box(A \rightarrow B)$ is a not theorem of KTB

1.	$\mathbb{F} 1, (\Box A \wedge \Box B) \rightarrow \Box\Box(A \rightarrow B)$ ✓	Assumption
2.	$\mathbb{T} 1, \Box A \wedge \Box B$ ✓	$\rightarrow\mathbb{F} 1$
3.	$\mathbb{F} 1, \Box\Box(A \rightarrow B)$ ✓	$\rightarrow\mathbb{F} 1$
4.	$\mathbb{T} 1, \Box A$	$\wedge\mathbb{T} 2$
5.	$\mathbb{T} 1, \Box B$	$\wedge\mathbb{T} 2$
6.	$\mathbb{T} 1, A$	$\mathbb{T}\Box 4$
7.	$\mathbb{T} 1, B$	$\mathbb{T}\Box 5$
8.	$\mathbb{F} 1.1, \Box(A \rightarrow B)$ ✓	$\Box\mathbb{F} 3$
9.	$\mathbb{T} 1.1, A$	$\Box\mathbb{T} 6$
10.	$\mathbb{T} 1.1, B$	$\Box\mathbb{T} 7$
11.	$\mathbb{F} 1.1.1, A \rightarrow B$ ✓	$\Box\mathbb{F} 8$
12.	$\mathbb{T} 1.1.1, A$	$\rightarrow\mathbb{F} 11$
13.	$\mathbb{F} 1.1.1, B$	$\rightarrow\mathbb{F} 11$

The tableau is open. The countermodel takes a bit more work to describe than the previous ones. There are three worlds, w_1 , $w_{1.1}$ and $w_{1.1.1}$. Every world can access every world *except*, w_1 and $w_{1.1.1}$ can access each other. (By the naming convention, $w_1 R w_{1.1}$ and $w_{1.1} R w_{1.1.1}$; adding reflexivity and symmetry get us the rest of the relations.) A is true at all three worlds, B is true at the first and second, but not the third.